CLASS 9 · ICSE · SELINA CONCISE MATHEMATICS · DPP SHEET

Chapter 28: Distance Formula

Daily practice problems: 10 questions with an answer key

Try every question on your own first. The answer key is on the last page.

1. Find the distance between $(1, 2)$ and $(4, 6)$.
2. Find the distance between $(-6, 7)$ and $(-1, -5)$.
3. Find the distance of $(8, -15)$ from the origin.
4. The distance between $(3, k)$ and $(0, 1)$ is 5. Find k.
5. Find the point on the x-axis equidistant from $(7, 6)$ and $(-3, 4)$.
6. Show that $(1, -1)$, $(5, 2)$ and $(9, 5)$ are collinear.
7. What kind of triangle has vertices $(0, 0)$, $(5, 0)$ and $(0, 12)$? Find its perimeter.
8. Show that $(2, 3)$, $(6, 3)$, $(6, 7)$ and $(2, 7)$ form a square, and find its area.
9. A circle with centre $(0, 0)$ passes through $(6, 8)$. Does $(10, 0)$ lie on it?
10. Find a relation between x and y if (x, y) is equidistant from $(2, 0)$ and $(0, 2)$.

MathSetu

Answer Key

1. 5 units
2. 13 units
3. 17 units
4. $k = 5$ or $k = -3$
5. $(3, 0)$
6. The distances are 5, 5 and 10, and $5 + 5 = 10$
7. Right-angled; $5 + 12 + 13 = 30$ units
8. Each side is 4 and both diagonals are $4\sqrt{2}$; area 16 square units
9. Yes; both are 10 units from the centre
10. $x = y$

MathSetu